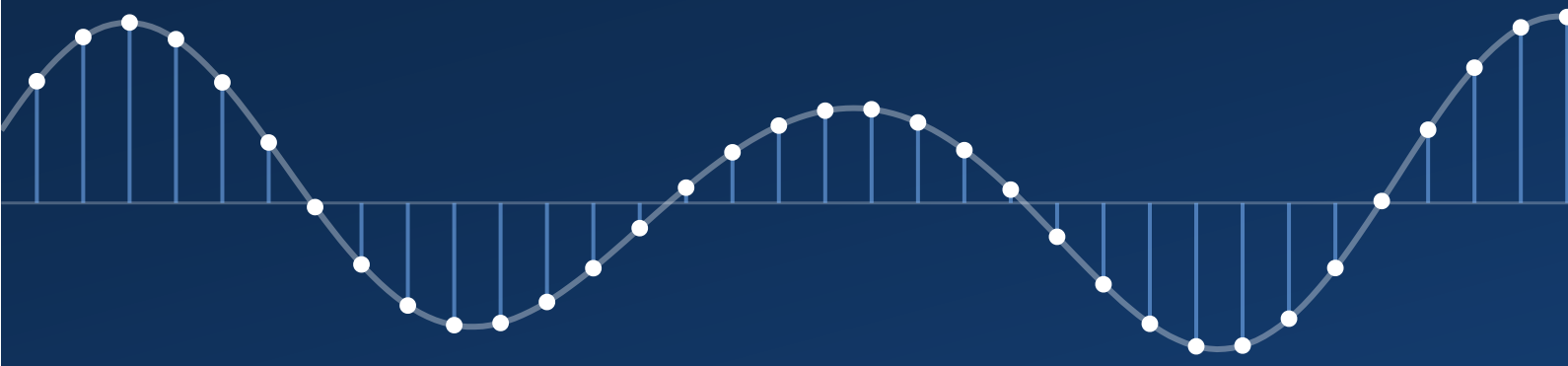
DIGITAL SIGNAL PROCESSING

CHAPTER 02

حل Sheet 2

حل نموذجي لكل أسئلة Sheet 2 (2.1 لحد 2.6)، بجُمل الدكتور من المحاضرة وبطريقته في الحل، زي ما يتكتب في ورقة الإجابة.

DFT · IDFT · Parseval · FFT · IFFT



6	1	5
أسئلة	نظري	مسائل

المحتوى

01 **الجزء النظري**
سؤال 2.1
1 سؤال 2.1 – 2.1

02 **المسائل**
أسئلة 2.2 لحد 2.6
5 سؤال 2.2 – 2.6

إزاي تستخدم الملف

الحل ماشي خطوة بخطوة بنفس طريقة الدكتور في المحاضرة والكتاب: الـ DFT زي Example 2.1، والـ IDFT زي Example 2.2، والـ FFT زي Example 2.3 و Fig. 2.3، والـ IFFT بنفس الـ flowgraph زي Sec. 2.6. السطر البرتقالي بيقولك الحل جاي من أنهي جزء في المحاضرة.

SECTION 01

الجزء النظري

سؤال 2.1

1. Prove that the DFT, X(k), of a sequence x(n) is periodic.

2.1 Prove that the DFT, X(k), of a sequence x(n) is periodic.

2.1 Prove that the DFT, X(k), of a sequence x(n) is periodic.

SOLUTION

An important property of the DFT may be deduced if the kth component of the DFT, X(k), is compared with the $(k + N)^{th}$ component, X(k + N). Thus:

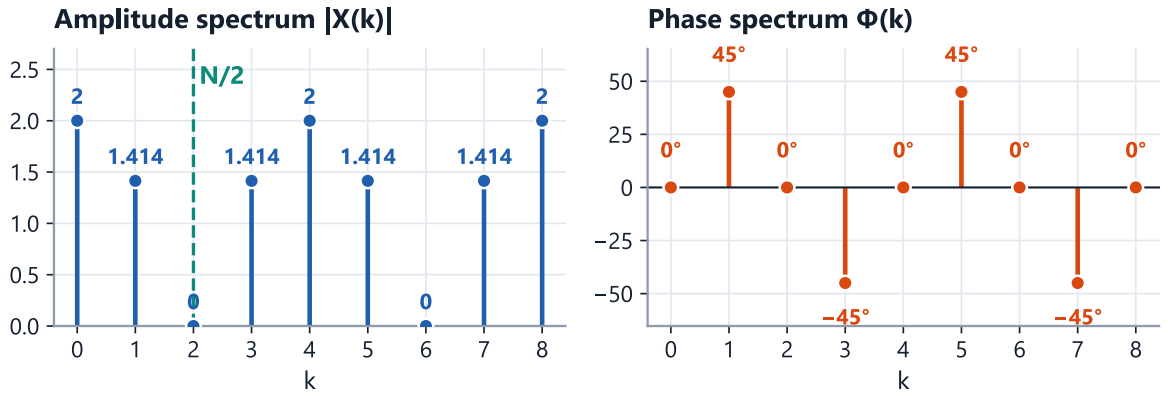
$$X(k) = \Sigma_{n=0}^{N-1} x(nT)\, e^{-jk\Omega nT} = \Sigma_{n=0}^{N-1} x(nT)\, e^{-j2\pi kn/N}$$

$$X(k + N) = \Sigma_{n=0}^{N-1} x(nT)\, e^{-j2\pi kn/N} \cdot e^{-j2\pi Nn/N} = \Sigma_{n=0}^{N-1} x(nT)\, e^{-j2\pi kn/N} \cdot e^{-j2\pi n} = X(k)$$

since n is integral, so $e^{-j2\pi n} = 1$.

The fact that **X(k + N) = X(k)** shows that the DFT is periodic with period N. This is the cyclical property of the DFT: the values of the DFT components are repetitive. If k = 0, then X(0) = X(N); e.g. for {1, 0, 0, 1}, X(0) = 2 and therefore X(4) = 2 also.

|X(k)| for {1, 0, 0, 1}: X(k + 4) = X(k), symmetric about N/2



Sec. 2.1 Lec 5 المرجع

SECTION 02

المسائل

أسئلة 2.2 لحد 2.6

1. DFT of {0, 1, 1, 0} and check by the IDFT

2. Verify Parseval's theorem for {0, 1, 1, 0}

3. FFT of {0, 1, 1, 0}, compare with the DFT, and IFFT

4. FFT of {0, 0, 1, 1, 1, 1, 0, 0}, spectra and IFFT

5. x(n) = cos(nπ/2): 8 samples, FFT and computational saving

2.2 DFT of {0, 1, 1, 0} and check by the IDFT

2.2 Calculate the DFT of the data sequence {0, 1, 1, 0} and check the validity of your answer by calculating its IDFT.

SOLUTION

DFT (as Example 2.1):

$$X(k) = \Sigma_{n=0}^{N-1} x(nT)\, e^{-jk\Omega nT} = \Sigma_{n=0}^{N-1} x(nT)\, e^{-j2\pi kn/N}, \quad \Omega = 2\pi/NT, \quad N = 4 \ (2.4)$$

x(0) = 0, x(T) = 1, x(2T) = 1, x(3T) = 0. It is required to find X(k) for k = 0, 1, 2, 3.

With k = 0:

$$\begin{aligned} X(0) &= \Sigma_{n=0}^{3} x(nT)\, e^{-j2\pi(0)n/4} \\ &= x(0) + x(T) + x(2T) + x(3T) \\ &= 0 + 1 + 1 + 0 \\ &= \mathbf{2} \end{aligned}$$

Thus X(0) = 2, of magnitude 2 and phase angle Φ(0) = 0°.

With k = 1:

$$\begin{aligned} X(1) &= \Sigma_{n=0}^{3} x(nT)\, e^{-j2\pi(1)n/4} \\ &= 0 + e^{-j\pi/2} + e^{-j\pi} + 0 \\ &= [\cos(\pi/2) - j\sin(\pi/2)] + [\cos(\pi) - j\sin(\pi)] \\ &= (-j) + (-1) \\ &= \mathbf{-1 - j} \end{aligned}$$

Thus X(1) = −1 − j, of magnitude √2 = 1.414 and phase angle Φ(1) = −135°.

With k = 2:

$$\begin{aligned} X(2) &= \Sigma_{n=0}^{3} x(nT)\, e^{-j2\pi(2)n/4} \\ &= 0 + e^{-j\pi} + e^{-j2\pi} + 0 \\ &= [\cos(\pi) - j\sin(\pi)] + [\cos(2\pi) - j\sin(2\pi)] \\ &= (-1) + 1 \\ &= \mathbf{0} \end{aligned}$$

Thus X(2) = 0, of magnitude 0 and phase angle Φ(2) = 0°.

With k = 3:

$$\begin{aligned} X(3) &= \Sigma_{n=0}^{3} x(nT)\, e^{-j2\pi(3)n/4} \\ &= 0 + e^{-j3\pi/2} + e^{-j3\pi} + 0 \end{aligned}$$

$$= [\cos(3\pi/2) - j\sin(3\pi/2)] + [\cos(3\pi) - j\sin(3\pi)]$$
$$= j + (-1)$$
$$= \mathbf{-1 + j}$$

Thus X(3) = −1 + j, of magnitude √2 = 1.414 and phase angle Φ(3) = 135°.

So, the time series x(n) = {0, 1, 1, 0} has the DFT given by **X(k) = {2, −1 − j, 0, −1 + j}**.

IDFT check (as Example 2.2):

$$x(nT) = F_D^{-1}[X(k)] = (1/N)\,\Sigma_{k=0}^{N-1} X(k)\, e^{jk\Omega nT} = (1/N)\,\Sigma_{k=0}^{N-1} X(k)\, e^{j2\pi kn/N} \quad (2.7)$$

With n = 0:

$$x(nT) = x(0) = (1/N)\,\Sigma_{k=0}^{3} X(k)\, e^{j2\pi k(0)/4}$$
$$= ¼\,[2 + (-1 - j) + 0 + (-1 + j)]$$
$$= \mathbf{0}, \text{ as expected.}$$

With n = 1:

$$x(nT) = x(T) = (1/N)\,\Sigma_{k=0}^{3} X(k)\, e^{j2\pi k(1)/4}$$
$$= ¼\,[2 + (-1 - j)\, e^{j\pi/2} + 0 + (-1 + j)\, e^{j3\pi/2}]$$
$$= ¼\,[2 + (-1 - j)(j) + (-1 + j)(-j)]$$
$$= ¼\,[2 + (1 - j) + (1 + j)]$$
$$= \mathbf{1}, \text{ as expected.}$$

With n = 2:

$$x(nT) = x(2T) = (1/N)\,\Sigma_{k=0}^{3} X(k)\, e^{j2\pi k(2)/4}$$
$$= ¼\,[2 + (-1 - j)\, e^{j\pi} + 0 + (-1 + j)\, e^{j3\pi}]$$
$$= ¼\,[2 + (-1 - j)(-1) + (-1 + j)(-1)]$$
$$= ¼\,[2 + (1 + j) + (1 - j)]$$
$$= \mathbf{1}, \text{ as expected.}$$

With n = 3:

$$x(nT) = x(3T) = (1/N)\,\Sigma_{k=0}^{3} X(k)\, e^{j2\pi k(3)/4}$$
$$= ¼\,[2 + (-1 - j)\, e^{j3\pi/2} + 0 + (-1 + j)\, e^{j9\pi/2}]$$
$$= ¼\,[2 + (-1 - j)(-j) + (-1 + j)(j)]$$
$$= ¼\,[2 + (-1 + j) + (-1 - j)]$$
$$= \mathbf{0}, \text{ as expected.}$$

∴ x(n) = **{0, 1, 1, 0}**, the original sequence.

المرجع Lec 5 Example 2.1, Example 2.2 Eq. 2.4, 2.7

2.3 Verify Parseval's theorem for {0, 1, 1, 0}

2.3 Use the results of prob. (2.2) to verify Parseval's theorem for the data {0, 1, 1, 0}.

SOLUTION

The normalized energy in the signal is given by either of the expressions:

$$\Sigma_{n=0}^{N-1} x^2(n) = (1/N) \Sigma_{k=0}^{N-1} |X(k)|^2 \quad (2.12)$$

$|X(k)| = \{2, \sqrt{2}, 0, \sqrt{2}\}$

$$\text{L.H.S.} = 0^2 + 1^2 + 1^2 + 0^2 = \mathbf{2}$$
$$\text{R.H.S.} = (1/4)[2^2 + \sqrt{2}^2 + 0^2 + \sqrt{2}^2] = (1/4)(8) = \mathbf{2}$$

L.H.S. = R.H.S., so Parseval's theorem is verified.

المرجع Lec 5 Sec. 2.3 (4) Eq. 2.12

2.4 FFT of {0, 1, 1, 0}, compare with the DFT, and IFFT

2.4 Calculate the DFT of the data sequence {0, 1, 1, 0} using the decimation-in-time FFT algorithm. Check the answer with that of Prob. (2.2). Compare numbers of complex additions and multiplications in the two methods. Calculate the IFFT to verify your answer.

SOLUTION

FFT (as Example 2.3):

This is a four-point DFT with x0 = 0, x1 = 1, x2 = 1, x3 = 0, and

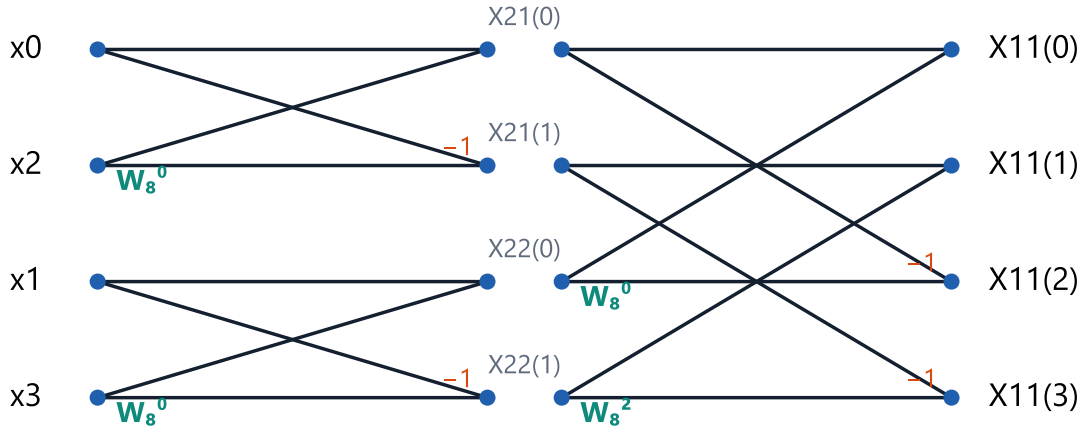
$$X1(k) = X11(k) + W_N^k X12(k), \quad k = 0, 1, 2, 3 \quad (2.24)$$

The re-ordered sequence is x0, x2, x1, x3. Using the top left-hand corner of Fig. 2.3 ($W_8^0 = 1$, $W_8^2 = e^{-j\pi/2} = -j$):

$$X21(0) = x0 + x2 = 0 + 1 = 1$$
$$X21(1) = x0 - x2 = 0 - 1 = -1$$
$$X22(0) = x1 + x3 = 1 + 0 = 1$$
$$X22(1) = x1 - x3 = 1 - 0 = 1$$

$$X11(0) = X21(0) + W_8^0\, X22(0) = 1 + 1 = \mathbf{2}$$
$$X11(1) = X21(1) + W_8^2\, X22(1) = -1 + e^{-j\pi/2}(1) = -1 - j(1) = \mathbf{-1 - j}$$
$$X11(2) = X21(0) - W_8^0\, X22(0) = 1 - 1 = \mathbf{0}$$
$$X11(3) = X21(1) - W_8^2\, X22(1) = -1 - e^{-j\pi/2}(1) = -1 + j(1) = \mathbf{-1 + j}$$

4-point DIT FFT (top left-hand corner of Fig. 2.3)



These values X(k) = {2, −1 − j, 0, −1 + j} are the same as those obtained in Prob. (2.2), but were more readily obtained using the FFT algorithm.

Comparison (Sec. 2.4, 2.5.2), N = 4:

Operation	DFT	FFT	Saving
No. of complex multiplications	$N^2 = 16$	$(N/2)\log_2 N = 4$	**12**
No. of complex additions	$N(N-1) = 12$	$N\log_2 N = 8$	**4**

IFFT (Sec. 2.6):

The IFFT (2.27) differs from the FFT (2.26) only in the factor 1/N and the sign of the exponent, so the same flowgraph is used with the re-ordered input X(0), X(2), X(1), X(3), $W_8^{-2} = e^{+j\pi/2} = +j$, and the result is divided by N = 4:

$$x21(0) = X(0) + X(2) = 2 \quad x21(1) = X(0) - X(2) = 2$$
$$x22(0) = X(1) + X(3) = -2 \quad x22(1) = X(1) - X(3) = -j2$$

$$x(0) = \tfrac{1}{4}[x21(0) + x22(0)] = \tfrac{1}{4}[0] = \mathbf{0}$$
$$x(1) = \tfrac{1}{4}[x21(1) + j\,x22(1)] = \tfrac{1}{4}[2 + j(-j2)] = \mathbf{1}$$
$$x(2) = \tfrac{1}{4}[x21(0) - x22(0)] = \tfrac{1}{4}[4] = \mathbf{1}$$
$$x(3) = \tfrac{1}{4}[x21(1) - j\,x22(1)] = \tfrac{1}{4}[2 - j(-j2)] = \mathbf{0}$$

∴ x(n) = **{0, 1, 1, 0}**, the original data sequence.

Sec. 2.6 Sec. 2.5.2 Example 2.3 Lec 6 **المرجع**

2.5 FFT of {0, 0, 1, 1, 1, 1, 0, 0}, spectra and IFFT

2.5 Calculate the FFT of the data sequence {0, 0, 1, 1, 1, 1, 0, 0} and plot the amplitude and phase spectra. Check your answer by calculating the IFFT to obtain the original data sequence.

SOLUTION

FFT (8-point, Fig. 2.3):

Re-ordered input (Table 2.1): x0, x4, x2, x6, x1, x5, x3, x7. $W_8^0 = 1$, $W_8^1 = (1 - j)/\sqrt{2}$, $W_8^2 = -j$, $W_8^3 = -(1 + j)/\sqrt{2}$

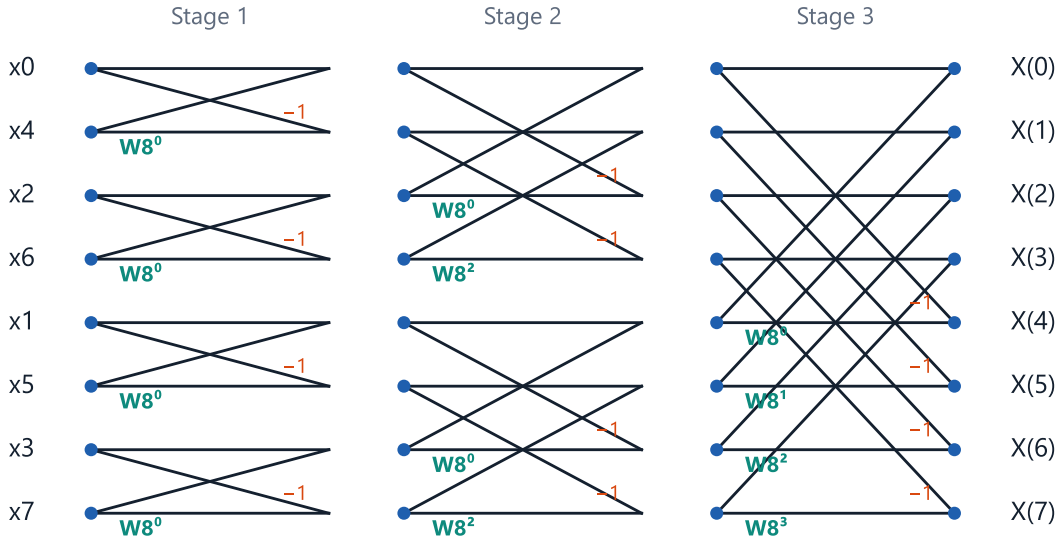
Stage 1: four 2-point DFTs	Stage 2: two 4-point DFTs
$X21(0) = x0 + x4 = 1$ $X21(1) = x0 - x4 = -1$	$X11(0) = X21(0) + W_8^0 X22(0) = 2$ $X11(1) = X21(1) + W_8^2 X22(1) = -1 - j$
$X22(0) = x2 + x6 = 1$ $X22(1) = x2 - x6 = 1$	$X11(2) = X21(0) - W_8^0 X22(0) = 0$ $X11(3) = X21(1) - W_8^2 X22(1) = -1 + j$
$X23(0) = x1 + x5 = 1$ $X23(1) = x1 - x5 = -1$	$X12(0) = X23(0) + W_8^0 X24(0) = 2$ $X12(1) = X23(1) + W_8^2 X24(1) = -1 - j$
$X24(0) = x3 + x7 = 1$ $X24(1) = x3 - x7 = 1$	$X12(2) = X23(0) - W_8^0 X24(0) = 0$ $X12(3) = X23(1) - W_8^2 X24(1) = -1 + j$

Stage 3: $X1(k) = X11(k) + W_8^k X12(k)$, $X1(k + 4) = X11(k) - W_8^k X12(k)$, $k = 0, 1, 2, 3$

$$X(0) = 4 \quad X(4) = 0$$
$$X(1) = -2.414 - j \quad X(5) = 0.414 - j$$
$$X(2) = 0 \quad X(6) = 0$$
$$X(3) = 0.414 + j \quad X(7) = -2.414 + j$$

$\therefore X(k) =$ **{4, −2.414 − j, 0, 0.414 + j, 0, 0.414 − j, 0, −2.414 + j}**

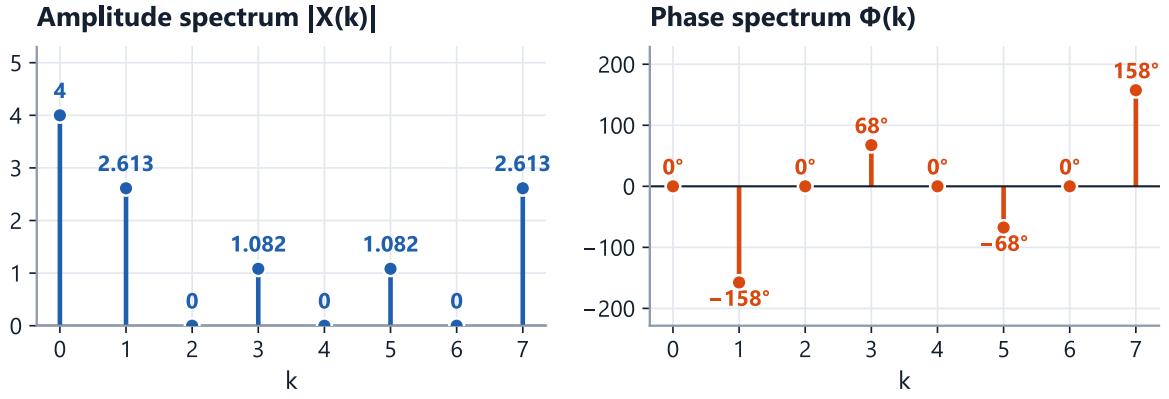
8-point DIT FFT: N/2 butterflies × $\log_2 N$ stages (Fig. 2.3)



Amplitude and phase spectra (|X(k)| and Φ(k) versus k; symmetrical about N/2 = 4):

k	0	1	2	3	4	5	6	7
X(k)	4	−2.414 − j	0	0.414 + j	0	0.414 − j	0	−2.414 + j
|X(k)|	4	2.613	0	1.082	0	1.082	0	2.613
Φ(k)	0°	−157.5°	0°	67.5°	0°	−67.5°	0°	157.5°

Spectra of {0, 0, 1, 1, 1, 1, 0, 0}



IFFT check (same flowgraph, W_8^{-k}, divided by N = 8):

Re-ordered input (Table 2.1): X(0), X(4), X(2), X(6), X(1), X(5), X(3), X(7). $W_8^0 = 1$, $W_8^{-1} = (1 + j)/\sqrt{2}$, $W_8^{-2} = +j$, $W_8^{-3} = (-1 + j)/\sqrt{2}$

Stage 1: four 2-point DFTs	Stage 2: two 4-point DFTs
$X21(0) = X(0) + X(4) = 4$ $X21(1) = X(0) - X(4) = 4$	$X11(0) = X21(0) + W_8^0 X22(0) = 4$ $X11(1) = X21(1) + W_8^{-2} X22(1) = 4$
$X22(0) = X(2) + X(6) = 0$ $X22(1) = X(2) - X(6) = 0$	$X11(2) = X21(0) - W_8^0 X22(0) = 4$ $X11(3) = X21(1) - W_8^{-2} X22(1) = 4$
$X23(0) = X(1) + X(5) = -2 - j2$ $X23(1) = X(1) - X(5) = -2.828$	$X12(0) = X23(0) + W_8^0 X24(0) = -4$ $X12(1) = X23(1) + W_8^{-2} X24(1) = -2.828 + j2.828$
$X24(0) = X(3) + X(7) = -2 + j2$ $X24(1) = X(3) - X(7) = 2.828$	$X12(2) = X23(0) - W_8^0 X24(0) = -j4$ $X12(3) = X23(1) - W_8^{-2} X24(1) = -2.828 - j2.828$

Stage 3: $X1(k) = X11(k) + W_8^{-k} X12(k), \quad X1(k+4) = X11(k) - W_8^{-k} X12(k), \quad k = 0, 1, 2, 3$

$x(0) = (1/8)(0) = \mathbf{0}$ $\quad x(4) = (1/8)(8) = \mathbf{1}$
$x(1) = (1/8)(0) = \mathbf{0}$ $\quad x(5) = (1/8)(8) = \mathbf{1}$
$x(2) = (1/8)(8) = \mathbf{1}$ $\quad x(6) = (1/8)(0) = \mathbf{0}$
$x(3) = (1/8)(8) = \mathbf{1}$ $\quad x(7) = (1/8)(0) = \mathbf{0}$

∴ $x(n) = \mathbf{\{0, 0, 1, 1, 1, 1, 0, 0\}}$, the original data sequence.

Sec. 2.6 Fig. 2.3 Table 2.1 Lec 6 **المرجع**

2.6 x(n) = cos(nπ/2): 8 samples, FFT and computational saving

2.6 For the signal $x(n) = \cos(n\pi/2)$, find the first 8 samples and then, using the FFT algorithm, calculate the frequency components X(k). Assume we use the DFT algorithm instead of the FFT algorithm, what is the computational saving?

SOLUTION

$x(n) = \cos(n\pi/2)$: $\quad x(0) = \cos 0 = 1, x(1) = \cos(\pi/2) = 0, x(2) = \cos \pi = -1, x(3) = \cos(3\pi/2) = 0, \ldots$

$x(n) = \mathbf{\{1, 0, -1, 0, 1, 0, -1, 0\}}$

FFT (8-point, Fig. 2.3):

Re-ordered input (Table 2.1): x0, x4, x2, x6, x1, x5, x3, x7. $\quad W_8^0 = 1, W_8^1 = (1 - j)/\sqrt{2}, W_8^2 = -j, W_8^3 = -(1 + j)/\sqrt{2}$

Stage 1: four 2-point DFTs	Stage 2: two 4-point DFTs
$X21(0) = x0 + x4 = 2$ $X21(1) = x0 - x4 = 0$	$X11(0) = X21(0) + W_8^0 X22(0) = 0$ $X11(1) = X21(1) + W_8^2 X22(1) = 0$
$X22(0) = x2 + x6 = -2$ $X22(1) = x2 - x6 = 0$	$X11(2) = X21(0) - W_8^0 X22(0) = 4$ $X11(3) = X21(1) - W_8^2 X22(1) = 0$
$X23(0) = x1 + x5 = 0$ $X23(1) = x1 - x5 = 0$	$X12(0) = X23(0) + W_8^0 X24(0) = 0$ $X12(1) = X23(1) + W_8^2 X24(1) = 0$
$X24(0) = x3 + x7 = 0$ $X24(1) = x3 - x7 = 0$	$X12(2) = X23(0) - W_8^0 X24(0) = 0$ $X12(3) = X23(1) - W_8^2 X24(1) = 0$

Stage 3: $X1(k) = X11(k) + W_8^k X12(k), \quad X1(k+4) = X11(k) - W_8^k X12(k), \quad k = 0, 1, 2, 3$

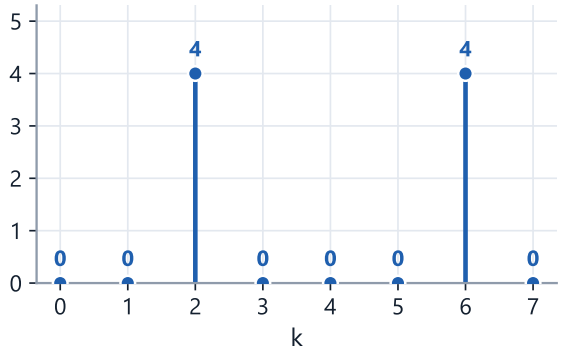
$$X(0) = 0 \quad X(4) = 0$$
$$X(1) = 0 \quad X(5) = 0$$
$$X(2) = 4 \quad X(6) = 4$$
$$X(3) = 0 \quad X(7) = 0$$

$\therefore X(k) = \mathbf{\{0, 0, 4, 0, 0, 0, 4, 0\}}$

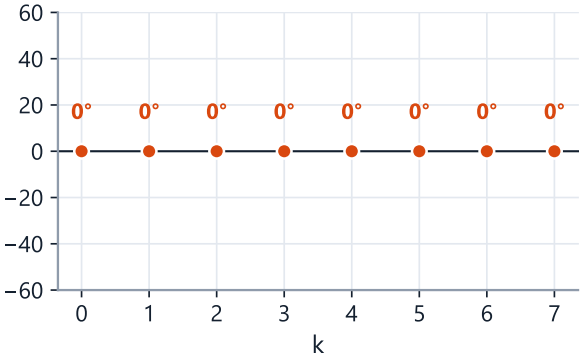
k	0	1	2	3	4	5	6	7
X(k)	0	0	4	0	0	0	4	0
|X(k)|	0	0	4	0	0	0	4	0
Φ(k)	0°	0°	0°	0°	0°	0°	0°	0°

Spectra of x(n) = cos(nπ/2)

Amplitude spectrum |X(k)|



Phase spectrum Φ(k)



Computational saving (Sec. 2.5.2), N = 8:

Operation	DFT	FFT	Saving
No. of complex multiplications	$N^2 = 64$	$(N/2)\log_2 N = 12$	**52**
No. of complex additions	$N(N-1) = 56$	$N \log_2 N = 24$	**32**

المرجع Lec 6 Fig. 2.3 Sec. 2.5.2 Table 2.2